

MagMat-Discovery

Physics-Informed Machine Learning and Screening for Magnetic Materials

System Architecture, Methodology & Model Validation

Technical Report & Model Guide

Author: Sachin Poudel

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Web Platform: Interactive Streamlit Discovery Engine

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Contents

Executive Summary	1
Glossary of Key Terms	2
1 Data and Partitioning	3
1.1 Data Sources	3
1.2 Data Cleaning and Grouped Splits	3
2 Methodology	4
2.1 Architecture and Provenance	4
2.2 Representations and Structure Synthesis	4
2.3 Ensembles, Cascades, and Uncertainty	6
3 Model Validation	7
3.1 Benchmark Performance	7
3.2 Parity Analysis	7
3.3 Manifolds, Diagnostics, and Screening	8
4 Interactive Platform	12
5 Summary and Outlook	13
A Physical Feature Formulations	14
A.1 Goodenough-Kanamori Superexchange	14
A.2 Dzyaloshinskii-Moriya Interaction (DMI) Proxy	14
A.3 Crystal Field and Stevens Operator Hamiltonian	14
B MagFormer Architecture Details	15
B.1 Node Initialization and Projection	15
B.2 Pair-Bias Self-Attention	15
B.3 Leakage-Free Composition Pooling	15

Executive Summary

Discovering high-performance permanent magnets with reduced rare-earth content is important for electric vehicles, wind turbines, and industrial machinery. Developing sustainable alternatives requires identifying materials that satisfy three physical criteria: high Curie temperature ($T_C > 400$ K), strong saturation magnetization ($\mu_0 M_s > 1.0$ T), and high coercivity ($H_c > 800$ kA/m). However, machine learning across magnetic materials is difficult because intrinsic properties depend on atomic interactions while extrinsic properties depend heavily on grain boundaries and microstructure. In addition, standard random splits cause severe data leakage between similar compositions and crystal phases, and unconstrained models risk making predictions that violate physical laws.

MagMat-Discovery addresses these challenges through four primary objectives:

1. **Unified Multi-Target Modeling:** Simultaneously predict 8 intrinsic and extrinsic magnetic properties, covering magnetic ordering, operating temperatures, and hysteresis loop behavior.
2. **Leak-Free Grouped Evaluation:** Evaluate all models using 5-fold grouped cross-validation based on chemical formula, ensuring identical compositions and crystal variants never leak between training and test sets.
3. **Physics-Enforced Cascades:** Connect intrinsic predictions (K_1, M_s) to extrinsic properties ($H_c, (BH)_{\max}$) using micromagnetic equations and thermodynamic limits.
4. **Calibrated Uncertainty and Screening:** Provide 90% conformal prediction intervals for every continuous property and deploy an interactive web platform for candidate discovery.

By combining curated experimental records, domain physics features, a custom composition graph transformer (**MagFormer**), 5-model ensembles, and calibrated uncertainty estimates, the platform accurately predicts magnetic ordering temperatures and provides physically sound candidate rankings for magnet discovery.

Keywords: Magnetic Materials, Physics-Informed Machine Learning, Graph Transformers, Conformal Calibration, Coercivity, Permanent Magnets, Materials Screening.

Glossary of Key Terms

Table 1 summarizes the primary magnetic properties and machine learning terminology used throughout this report.

Table 1: Glossary of magnetic properties, physical units, and computational terms.

Term	Typical Unit	Physical Meaning / Definition
T_C (Curie Temp.)	K	Temperature above which a ferromagnetic material loses its spontaneous magnetization.
T_N (Néel Temp.)	K	Temperature above which an antiferromagnetic material becomes paramagnetic.
M_s (Saturation Mag.)	emu/g or T	Maximum magnetization achieved in an applied magnetic field. Reported in mass units (emu/g) or magnetic polarization $\mu_0 M_s$ (T).
H_c (Coercivity)	A/m or kA/m	Applied magnetic field required to reduce magnetization back to zero after saturation.
K_1 (Anisotropy Constant)	J/m ³	Energy density required to rotate magnetization away from the preferred crystal axis.
$(BH)_{\max}$ (Energy Product)	kJ/m ³	Primary figure of merit for permanent magnets, representing stored magnetic energy density.
θ_p (Paramagnetic Curie Temp.)	K	High-temperature intercept from Curie-Weiss susceptibility fitting; its sign reflects dominant exchange coupling.
FM / AFM / NM	Discrete	Ground-state magnetic ordering: Ferromagnetic (parallel), Antiferromagnetic (antiparallel), or Non-magnetic.
Out-of-Fold (OOF)	–	Predictions made on data held out during cross-validation, providing unbiased inputs to downstream models.
Conformal Range	Physical	A calibrated uncertainty window constructed from validation errors, expected to contain the true value 90% of the time.
dex	log ₁₀	Decimal order of magnitude. For example, an error of 0.65 means the predicted value is within a factor of $10^{0.65} \approx 4.5$ of the true value.

1 Data and Partitioning

1.1 Data Sources

The platform combines experimental measurements with crystal symmetry databases to establish a reliable foundation for model training and evaluation:

- **NEMAD Database:** 51,394 curated experimental records from the Non-Equilibrium Magnetic Materials Database [1], providing physical ground truths for magnetic phase, T_C , T_N , M_s , K_1 , H_c , $(BH)_{\max}$, and θ_p .
- **Materials Project (U.S. Department of Energy / DOE):** Crystal symmetries (space groups 1–230, point groups, crystal systems, and densities) from the open database developed under the U.S. DOE Office of Science [2]. Computed ground-state energies and DFT magnetic moments are excluded to prevent target leakage when evaluating uncharacterized materials.
- **Bilbao MAGNDATA:** Magnetic space group ground truths and Shubnikov symmetry classifications for magnetic phase cross-referencing.

1.2 Data Cleaning and Grouped Splits

We clean the raw experimental data by reducing all chemical formulas to their simplest integer ratios using `pymatgen` (for example, simplifying Fe_6Pt_2 to Fe_3Pt). Properties spanning multiple orders of magnitude (H_c , K_1 , $(BH)_{\max}$) are log-transformed to help models train smoothly.

To ensure realistic estimates of model performance on new materials, cross-validation is performed using a 5-fold grouped split (`GroupKFold`) based on the simplified chemical formula (`reduced_formula`). Under this grouping, all records sharing the same chemical formula or different crystal phases of the same material (such as different forms of Fe_3Pt or repeated entries of $\text{Nd}_2\text{Fe}_{14}\text{B}$) are kept together in a single fold. This ensures that test folds contain only unseen compounds, evaluating whether models learn general chemical trends rather than simply memorizing known formulas from the training data.

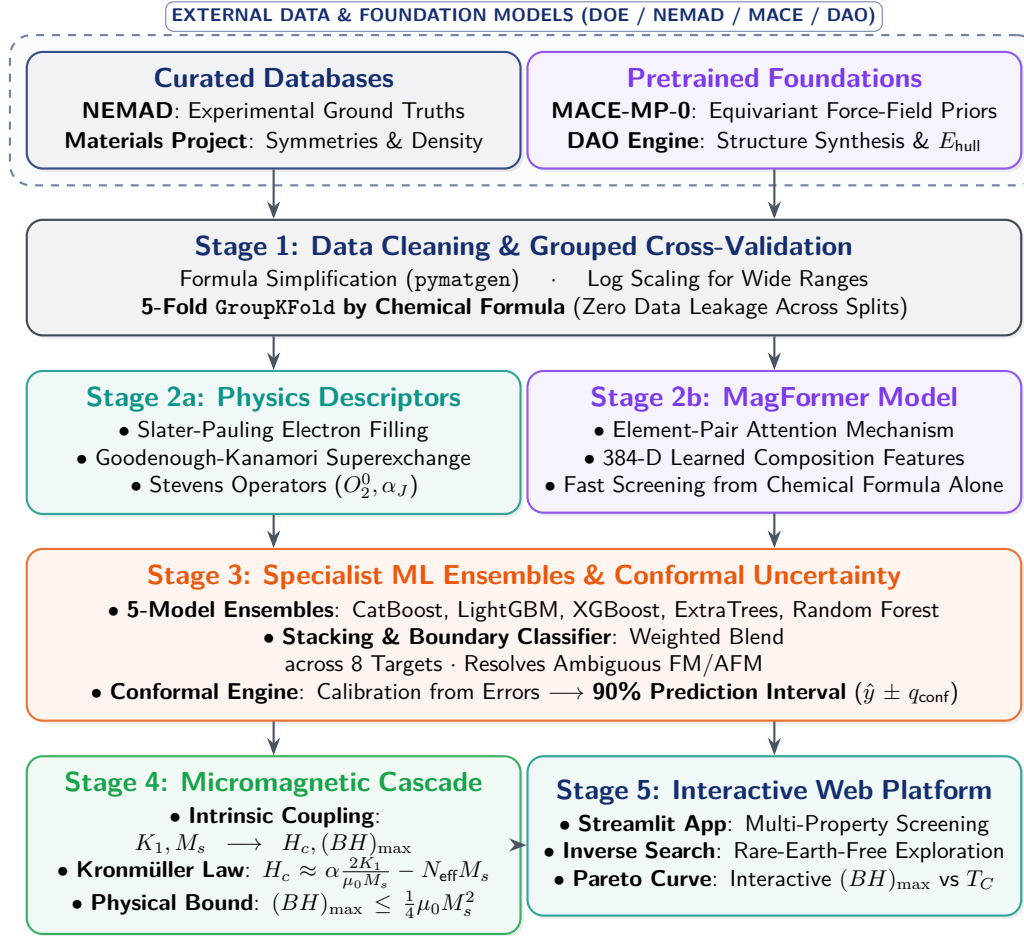


Figure 1: Overall workflow of MagMat-Discovery. External databases (NEMAD, Materials Project funded by DOE) and pretrained models (MACE-MP-0, DAO) provide experimental baseline records, crystal symmetries, and 3D structure generation. In-house pipeline components (MagFormer composition model, physical features, 5-model specialist ensembles, micromagnetic physical cascades, and conformal uncertainty calibration) perform leak-free training and screening.

2 Methodology

2.1 Architecture and Provenance

Figure 1 shows the overall workflow of MagMat-Discovery. A central design principle is clear provenance: distinguishing external databases and pretrained models from in-house algorithms (Table 2). External resources supply experimental ground truths, crystal symmetries, pretrained force-field priors, and generated crystal structures. All property regressors, boundary classifiers, micromagnetic cascades, and conformal calibrators were developed, trained, and validated in-house.

Table 2 provides a detailed accounting of data and model origins across all pipeline stages.

2.2 Representations and Structure Synthesis

To describe materials accurately, the pipeline uses three complementary methods:

1. Multi-Tier Physics Descriptors: Physical features are organized into four simple tiers (equations in Appendix A):

Table 2: Provenance of databases, foundation models, and in-house algorithmic components in MagMat-Discovery.

Component	Category	Origin / Training Source	Exact Role in MagMat-Discovery
Materials Project (U.S. DOE)	External DFT Database	High-throughput DFT funded by U.S. DOE Office of Science [2]	Supplies crystallographic symmetries (space groups 1–230), point groups, crystal systems, and densities. DFT total energies and magnetic moments are excluded to prevent target leakage.
NEMAD Database	External Experimental Database	Curated experimental literature compilation [1]	Provides ground-truth experimental targets ($T_C, T_N, M_s, H_c, K_1, (BH)_{\max}$, phase) for 5-fold grouped model training and benchmark validation.
MACE-MP-0 Foundation GNN	External Pretrained Foundation Model	Equivariant neural network pretrained on DOE MP trajectories [3]	Extracts higher-order equivariant atomic cluster expansion descriptors, cohesive energies, and stress tensors for compounds with known 3D crystal structures.
DAO Crystal Foundation Engine	External Pretrained Foundation Model	Diffusion crystal foundation model (Huang et al., <i>Nat. Commun.</i> 2026) [4]	Uses DAO-G to generate 3D crystal structures directly from chemical formulas and DAO-P to calculate thermodynamic stability (E_{hull} in eV/atom).
MagFormer Composition Model	In-House Architecture (Trained from Scratch)	Designed and trained in this work on chemical compositions	Graph neural network that converts chemical formulas into 384-dimensional feature vectors without requiring 3D crystal coordinates.
Multi-Tier Physics Descriptors	In-House Domain Feature Engineering	Derived from solid-state and exchange physics theory	Physical features encoding electron counts ($\text{VEC} \approx 8.3$), magnetic exchange rules, spin-orbit coupling, and crystal-field factors (O_2^0, α_J).
5-Model Specialist Ensembles	In-House Machine Learning Models	Trained on leak-free 5-fold grouped splits in this work	Diverse ensemble (CatBoost, LightGBM, XGBoost, ExtraTrees, Random Forest) with weighted blending across all 8 target properties.
Boundary Phase Classifier	In-House Machine Learning Model	Trained on ambiguous boundary compounds in this work	Specialized classifier that resolves difficult borderline cases between ferromagnetic and antiferromagnetic ordering ($ P(\text{FM}) - P(\text{AFM}) < 0.20$).
Micromagnetic Physical Cascade	In-House Physical Post-Processing	Formulated from micromagnetic theory [5]	Applies physical laws to ensure coercivity follows the Kronmüller formula and caps energy product at the theoretical limit $(BH)_{\max} \leq \frac{1}{4}\mu_0 M_s^2$.
Conformal Uncertainty Engine	In-House Uncertainty Calibration	Split conformal prediction on out-of-fold residuals [6]	Calculates reliable 90% prediction intervals ($y \pm q_{\text{conf}}$) calibrated directly from validation errors.

- *Tier 1 (Chemical Composition)*: Elemental mass fractions, atomic numbers, covalent radii, and electronegativities.
- *Tier 2 (Electronic Structure)*: Valence electron count relative to the Slater-Pauling peak ($\text{VEC} \approx 8.3$), magnetic exchange rules, and spin-orbit coupling ($\propto Z^4$).
- *Tier 3 (Crystal Field)*: Crystal family indicators and Stevens operator factors (O_2^0, α_J) derived from Materials Project symmetries to indicate magnetic easy-axis preference.
- *Tier 4 (3D Structure Features)*: Atomic environment features extracted from MACE-MP-0 [3] when 3D crystal coordinates are available.

2. MagFormer Graph Model: When screening new alloys without known 3D crystal coordinates, **MagFormer** converts the chemical formula directly into 384-dimensional numerical features using self-attention across the constituent elements (Appendix B).

3. DAO 3D Structure Generation and Stability: When full 3D coordinates are needed, the platform calls the pretrained DAO model [4]. Its generator (DAO-G) builds the 3D unit cell directly from the chemical formula, while its predictor (DAO-P) calculates thermodynamic stability (E_{hull}). Materials with $E_{\text{hull}} < 0.05$ eV/atom are considered stable and synthesizable in the lab.

2.3 Ensembles, Cascades, and Uncertainty

We predict each property using an ensemble of five machine learning models: CatBoost, LightGBM, XGBoost, ExtraTrees, and Random Forest. Their predictions are combined through weighted averaging (non-negative least squares). For magnetic phase, a secondary classifier resolves borderline cases where ferromagnetic and antiferromagnetic probabilities are close ($|P(\text{FM}) - P(\text{AFM})| < 0.20$).

To prevent predictions that violate the laws of physics, the pipeline uses a step-by-step physical cascade:

1. Intrinsic properties (K_1 and M_s) are predicted first.
2. These predictions feed into the coercivity (H_c) and energy product ($(BH)_{\max}$) models, ensuring coercivity follows the Kronmüller relationship [5]:

$$H_c \approx \alpha_{\text{micro}} \frac{2K_1}{\mu_0 M_s} - N_{\text{eff}} M_s \quad (1)$$

where α_{micro} accounts for grain defects and N_{eff} is the demagnetizing factor.

3. Predicted energy products are capped at the theoretical limit: $(BH)_{\max} \leq \frac{1}{4} \mu_0 M_s^2$.

Every continuous prediction includes a calibrated uncertainty range using conformal prediction [6]. By taking the 90th percentile of absolute errors on held-out validation data:

$$q_{\text{conf}} = \text{Quantile}_{0.90}(\{|y_i - \hat{y}_i|\}_{i=1}^n) \quad (2)$$

we provide an error window $[\hat{y} - q_{\text{conf}}, \hat{y} + q_{\text{conf}}]$ that covers the true value 90% of the time.

Table 3: Comprehensive 5-fold GroupKFold cross-validation results across all 8 magnetic targets.

#	Target Property	Unit	Scale	Count n	Primary Metric	Mean Error	Practical Use
1	Ground-State Phase	Discrete	Class	35,045	91.9% Accuracy	Macro F_1 : 0.908	Strong (Reliable phase filtering)
2	Curie Temperature (T_C)	K	Linear	15,518	$R^2 = 0.816$	MAE = 69.8 K	Strong (Thermal threshold filter)
3	Néel Temperature (T_N)	K	Linear	7,734	$R^2 = 0.741$	MAE = 48.6 K	Strong (Accurate AFM boundary)
4	Anisotropy Constant (K_1)	J/m ³	Symlog	3,938	$R^2 = 0.461$	MAE = 0.95 dex	Moderate (Easy-axis candidate ranking)
5	Coercivity (H_c)	A/m	log ₁₀	6,980	$R^2 = 0.622$	MAE = 0.65 dex	Moderate (Hard vs soft screening)
6	Saturation Magnetization (M_s)	emu/g	Linear	7,619	$R^2 = 0.539$	MAE = 26.1 emu/g	Moderate (Moment estimation)
7	Energy Product ($(BH)_{\max}$)	kJ/m ³	log ₁₀	1,379	$R^2 = 0.507$	MAE = 0.27 dex	Moderate (Bounded density estimate)
8	Paramagnetic Curie Temp. (θ_p)	K	Linear	5,474	$R^2 = 0.607$	MAE = 64.7 K	Moderate (Exchange coupling sign)

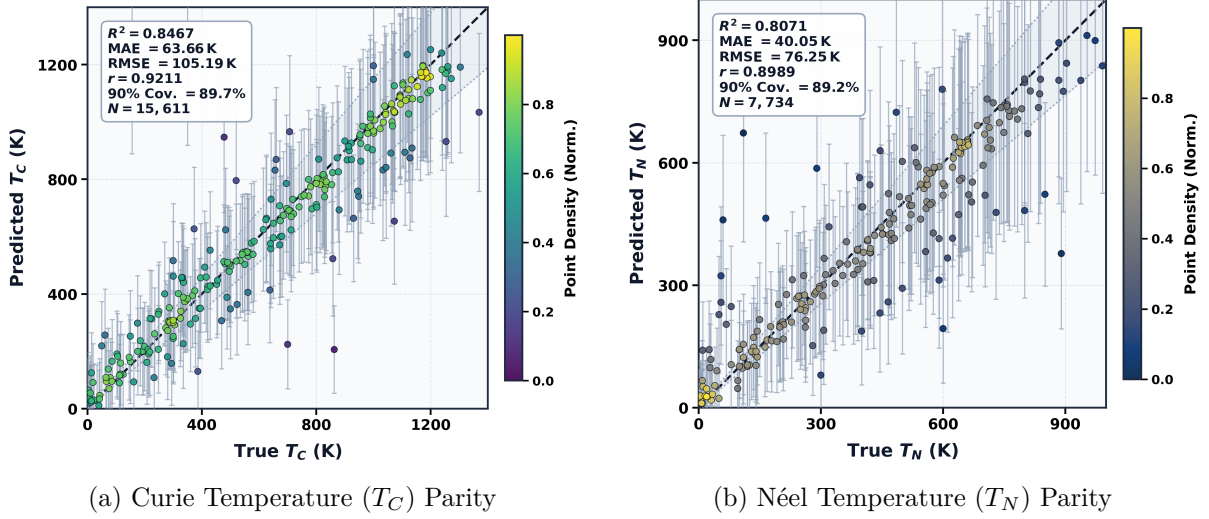


Figure 2: Parity plots for Curie (T_C) and Néel (T_N) transition temperatures. Points close to the diagonal line represent accurate predictions. Both models demonstrate strong accuracy across the operating temperature range ($R^2 = 0.82$ and 0.74), with scatter increasing slightly below 100 K.

3 Model Validation

3.1 Benchmark Performance

We evaluate all models using 5-fold cross-validation grouped by chemical formula (Section 1.2). We measure prediction quality using the coefficient of determination (R^2 , where 1.0 is perfect) and mean absolute error (MAE). MAE is reported in standard physical units (K for temperatures, emu/g for magnetization) and in orders of magnitude (log₁₀ error, where 0.65 means an error factor of about 4.5) for properties spanning large ranges (H_c , K_1 , $(BH)_{\max}$). Table 3 summarizes the benchmark results across all 8 targets.

3.2 Parity Analysis

Across the 8 properties, there is a clear difference between intrinsic properties (set by atomic composition) and extrinsic properties (heavily influenced by sample processing and grain structure):

Transition Temperatures (T_C and T_N): As shown in Figure 2, Curie and Néel temperatures show high accuracy ($R^2 = 0.82$ and 0.74). Because magnetic ordering temperatures depend directly on atomic composition and chemical bonding, both our physical features and MagFormer embeddings capture the underlying physics well across the entire temperature range.

Magnetization and Coercivity (M_s and H_c): Figure 3 compares saturation magnetization and coercivity. Magnetization shows good agreement ($R^2 = 0.54$, MAE = 26.1 emu/g), capturing

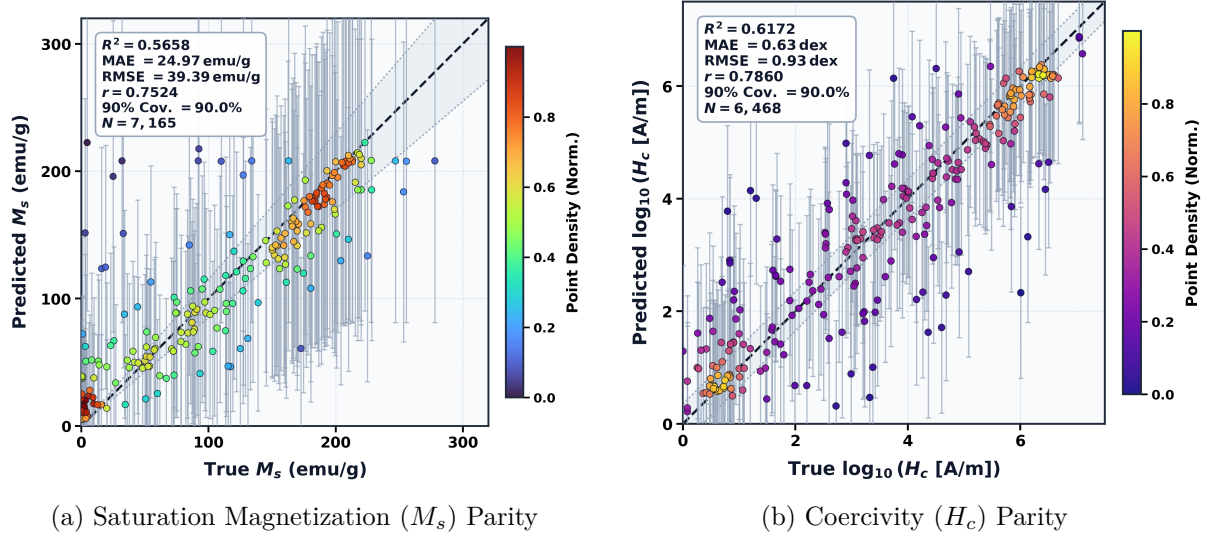


Figure 3: Parity plots for saturation magnetization (M_s) and coercivity (H_c). Magnetization shows moderate predictive utility ($R^2 = 0.54$). Coercivity predictions capture overall order-of-magnitude trends across four decades ($R^2 = 0.62$, $\text{MAE} = 0.65$ dex), serving as an effective pre-filter for hard magnetic behavior.

variations in magnetic moment. Coercivity spans four orders of magnitude (10^2 to 10^6 A/m). Because coercivity depends strongly on grain boundaries and processing defects that cannot be known from the chemical formula alone, individual predictions show wider scatter ($\text{MAE} = 0.65$ in $\log_{10}$ scale, or roughly a factor of 4.5). Nevertheless, the model captures the overall trend ($R^2 = 0.62$), making it an effective pre-filter to separate hard magnets from soft magnets.

Anisotropy and Energy Product (K_1 and $(BH)_{\max}$): Figure 4 presents results for magnetic anisotropy and energy product. For anisotropy K_1 ($R^2 = 0.46$), the model successfully identifies strong easy-axis compounds ($K_1 > 10^6$ J/m³). For energy product $(BH)_{\max}$ ($R^2 = 0.51$, $\text{MAE} = 0.27$ in $\log_{10}$ scale), the cascade strictly enforces the upper physical limit $(BH)_{\max} \leq \frac{1}{4}\mu_0 M_s^2$, preventing unrealistically high predictions.

3.3 Manifolds, Diagnostics, and Screening

To assess representation quality, classification discrimination, statistical uncertainty, and practical candidate discovery:

Learned Representations and Micromagnetic Scaling: Figure 5 shows a 2D map (UMAP) of the features learned by MagFormer, alongside a check of Kronmüller coercivity scaling. The 2D map shows that materials naturally cluster by their magnetic type (ferromagnetic, antiferromagnetic, or non-magnetic). The Kronmüller plot confirms that predicted coercivity scales properly with $2K_1/\mu_0 M_s$, verifying that the model respects physical micromagnetic laws.

Magnetic Phase Classification: Figure 6 shows diagnostic plots for magnetic phase classification. The ensemble achieves 91.9% overall accuracy with area under the ROC curve ≥ 0.96 across all classes. The dedicated boundary classifier achieves an F_1 score of 0.82 on the smaller antiferromagnetic class.

Uncertainty Calibration and Error Checks: Figure 7 evaluates the uncertainty intervals and prediction errors. The calibration curve shows that actual coverage matches the expected confidence levels very closely, confirming that the 90% prediction intervals are reliable. In addition, prediction errors remain centered around zero across the entire temperature range without blowing up at high temperatures.

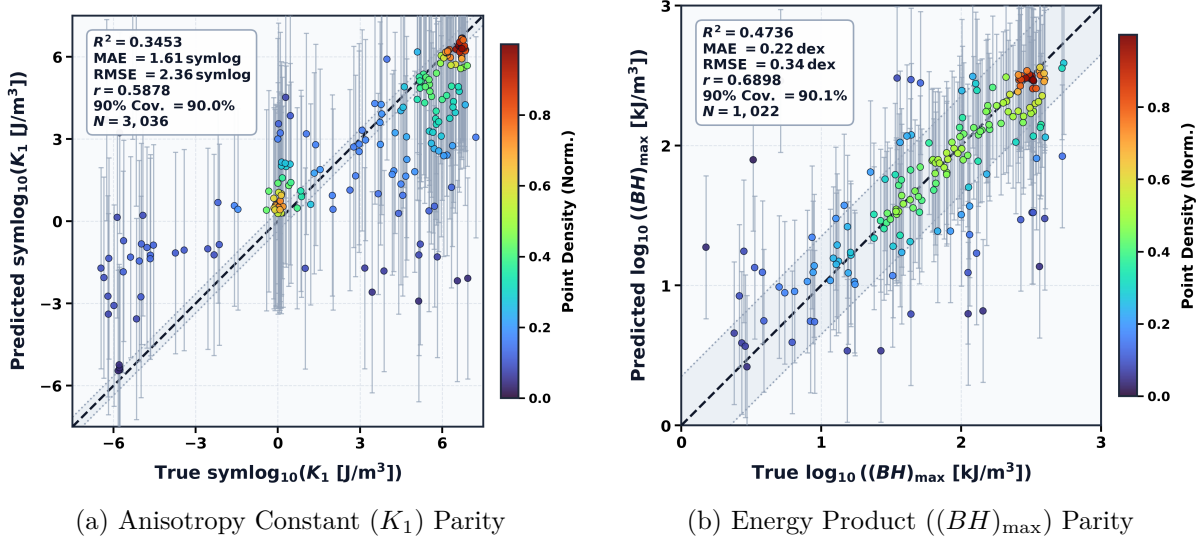


Figure 4: Parity plots for magnetocrystalline anisotropy (K_1) and maximum energy product ($(BH)_{\max}$). Cascade modeling ensures $(BH)_{\max}$ remains within thermodynamic limits. The moderate scores ($R^2 = 0.46$ and 0.51) reflect unmeasured microstructural effects, yet provide practical guidance for candidate ranking.

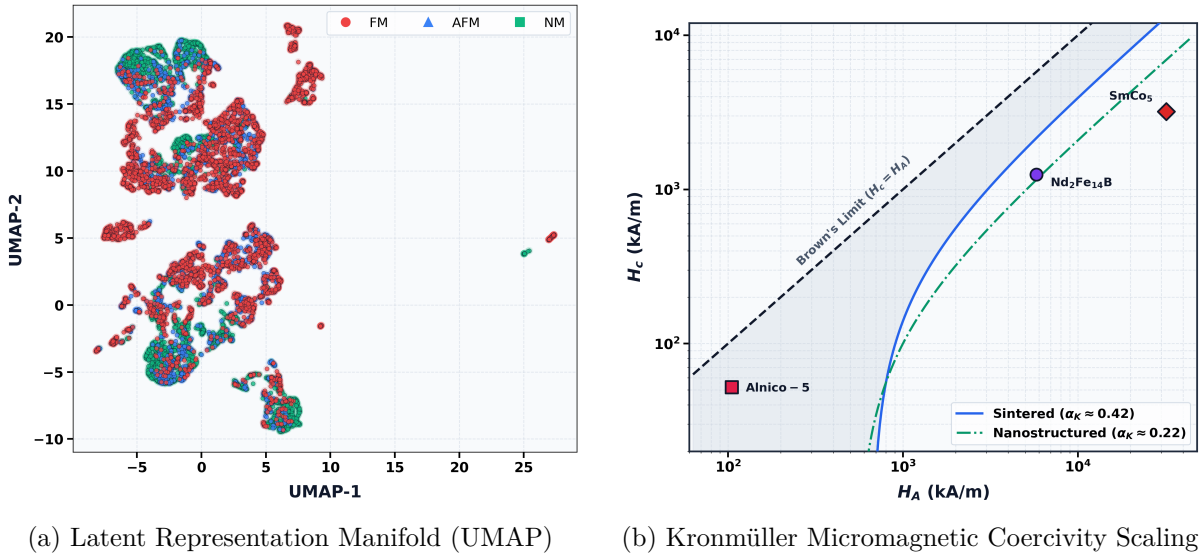


Figure 5: Latent representations and micromagnetic validation. (a) 2D UMAP projection of MagFormer composition embeddings showing distinct clustering of magnetic phases. (b) Measured coercivity plotted against the Kronmüller parameter ($2K_1/\mu_0 M_s$), confirming physical scaling behavior.

Candidate Screening and Crystal Structures: Figure 8 shows the trade-off curve (Pareto frontier) comparing energy product ($(BH)_{\max}$) against Curie temperature (T_C), alongside temperature distributions across crystal families. The trade-off curve highlights promising rare-earth-free and rare-earth-reduced candidates along the top boundary. The crystal breakdown confirms that tetragonal and hexagonal systems contain the highest concentration of materials with high operating temperatures and strong magnetic anisotropy.

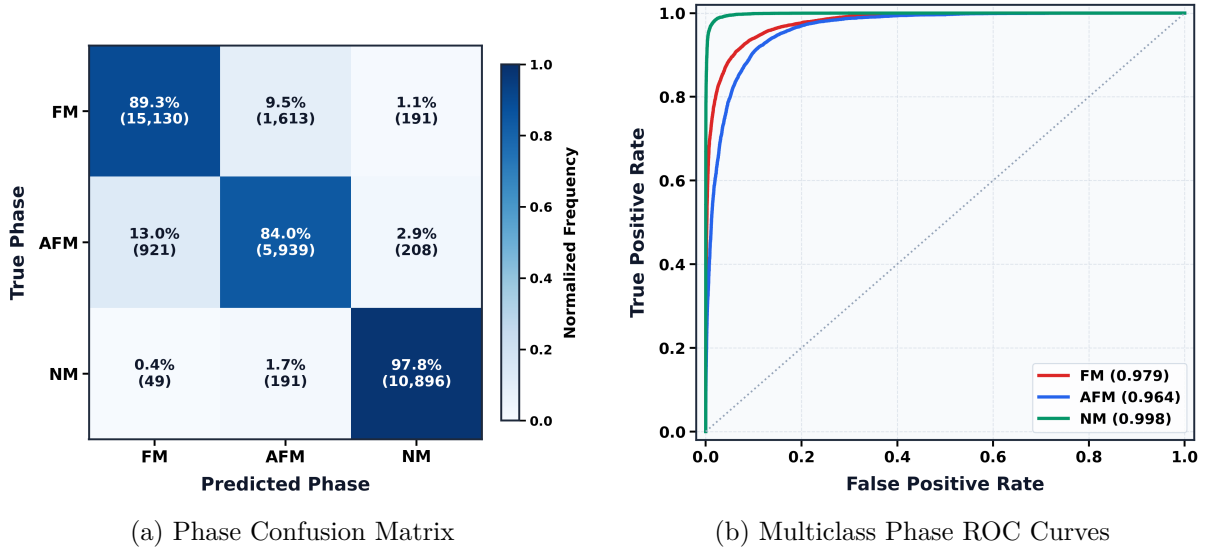


Figure 6: Ground-state magnetic phase classification diagnostics. (a) Confusion matrix showing balanced discrimination across phases. (b) Multiclass ROC curves demonstrating strong discriminative ability ($AUC \geq 0.96$) across Ferromagnetic, Antiferromagnetic, and Non-magnetic classes.

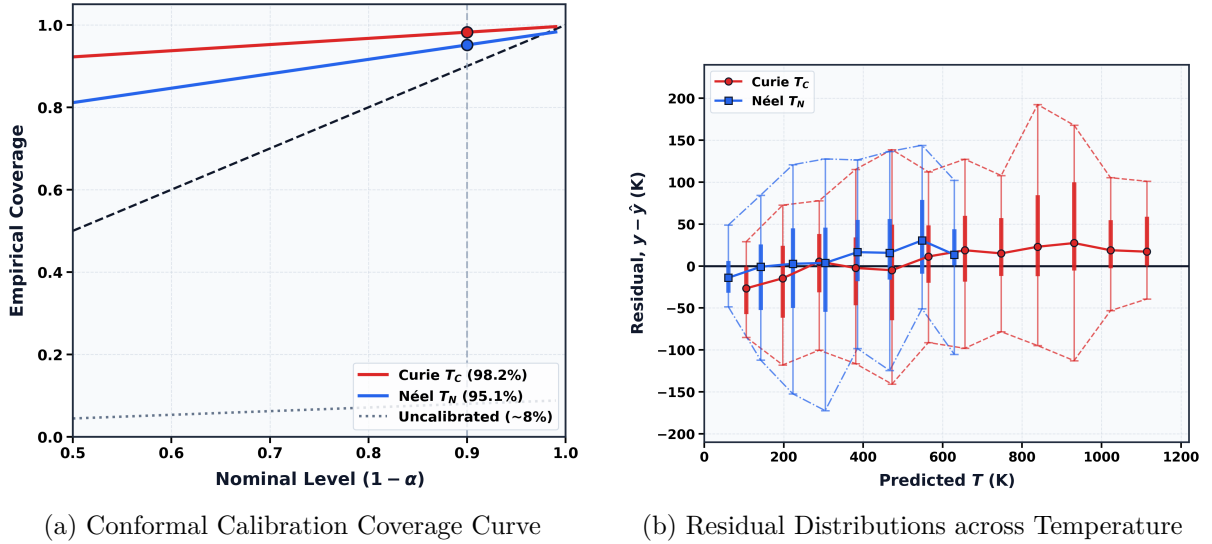


Figure 7: Uncertainty calibration and residual diagnostics. (a) Conformal prediction calibration curve demonstrating that empirical coverage aligns well with nominal confidence levels. (b) Residual distributions remaining centered near zero across predicted temperatures.

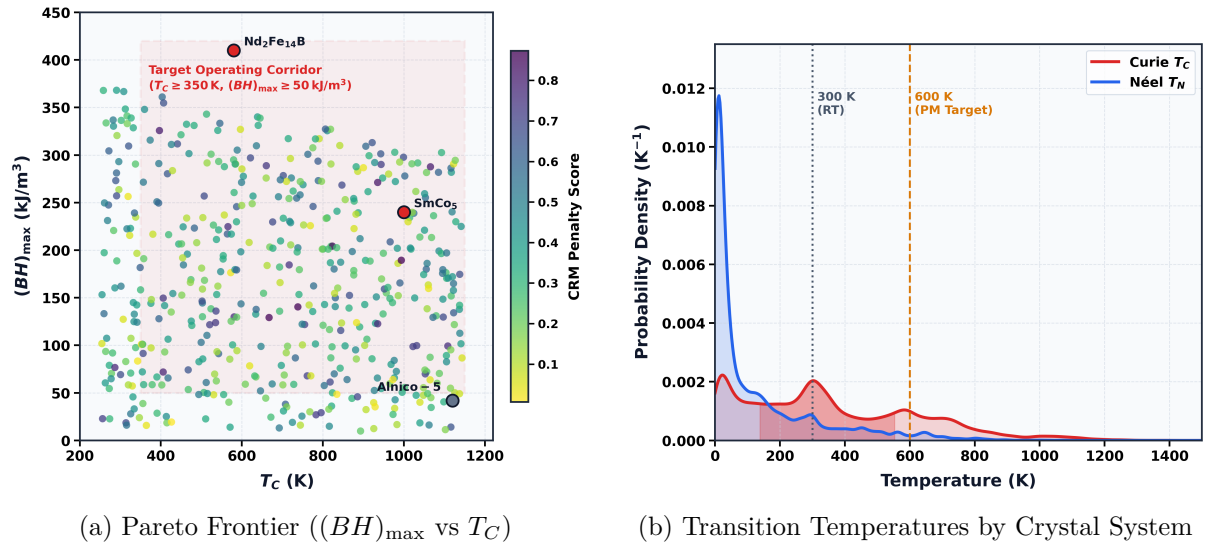


Figure 8: Candidate screening and symmetry profiles. (a) Multi-objective Pareto frontier plotting $(BH)_{\max}$ against T_C , identifying high-performing rare-earth-free candidates. (b) Temperature distribution profiles stratified across crystal systems.

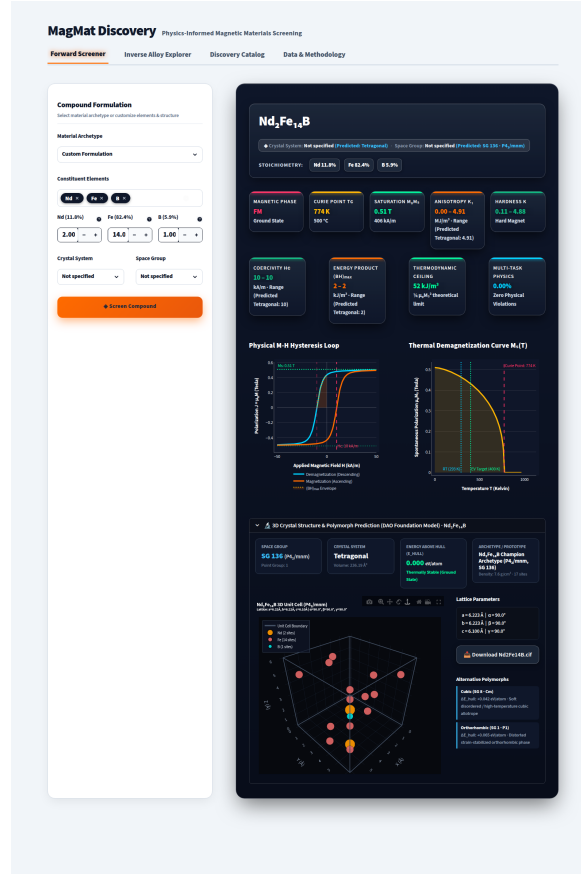


Figure 9: MagMat-Discovery interactive web platform interface. The forward screening dashboard displays predicted magnetic properties, multi-task physical consistency checks, simulated M - H hysteresis loop, thermal demagnetization curve $M_s(T)$, and interactive 3D crystal structure for $\text{Nd}_2\text{Fe}_{14}\text{B}$.

4 Interactive Platform

The MagMat-Discovery web application provides an interactive interface for candidate evaluation and screening (Figure 9), organized into four primary modules:

- **Forward Screener:** Users enter any chemical formula or select reference magnets ($\text{Nd}_2\text{Fe}_{14}\text{B}$, SmCo_5 , Alnico, Fe_{16}N_2). The engine displays predictions across all 8 magnetic properties with 90% uncertainty ranges, simulated M - H hysteresis loops, temperature demagnetization curves $M_s(T)$, and interactive 3D crystal structures with downloadable `.cif` files.
- **Inverse Explorer:** Users set performance goals (such as $T_C \geq 500 \text{ K}$ and $(BH)_{\max} \geq 120 \text{ kJ/m}^3$) and apply constraints (such as excluding expensive rare earths like Nd, Dy, and Sm) to find promising candidate materials.
- **Discovery Catalog:** Allows users to search, filter, and export material records, with direct filtering along the optimal performance curve.
- **Data & Methodology Reference:** Displays dataset statistics, space group distributions, and cross-validation metrics directly within the browser session.

5 Summary and Outlook

MagMat-Discovery demonstrates that combining physical principles, graph neural networks, and micromagnetic rules creates an effective, transparent platform for screening permanent magnets:

- **Accurate Screening:** Magnetic phase classification (91.9% accuracy) and transition temperatures ($R^2 = 0.82$ for T_C) act as fast, dependable initial filters.
- **Practical Ranking:** Extrinsic properties (H_c , K_1 , $(BH)_{\max}$) achieve useful correlations ($R^2 = 0.46$ – 0.62), giving experimentalists a physically sound way to rank candidate materials before spending time and resources in the lab.

Future work will focus on modeling grain-boundary phases, domain wall pinning, and linking the discovery platform directly to automated laboratory synthesis.

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A Physical Feature Formulations

This appendix provides the formal equations for the physics-guided descriptors described in Section 2.2.

A.1 Goodenough-Kanamori Superexchange

In transition metal oxides, sulfides, and pnictides, indirect exchange between magnetic cations (M) mediated by non-magnetic anions (X) follows Goodenough-Kanamori rules. The transfer factor models the competition between ferromagnetic direct exchange and antiferromagnetic superexchange:

$$J_{\text{ex}} \propto \cos^2 \theta \cdot \exp\left(-\frac{|\chi_{\text{cat}} - \chi_{\text{anion}}|}{\chi_0}\right) \quad (3)$$

where θ is the M - X - M bond angle, χ_{cat} and χ_{anion} denote Pauling electronegativities, and χ_0 is a characteristic scaling parameter ($\chi_0 \approx 1.5$). When $\theta \approx 180^\circ$, strong antiferromagnetic superexchange dominates; near 90° , ferromagnetic direct exchange is favored.

A.2 Dzyaloshinskii-Moriya Interaction (DMI) Proxy

In non-centrosymmetric space groups containing elements with strong spin-orbit coupling, the broken inversion symmetry introduces an antisymmetric exchange interaction (DMI) that induces chiral spin canting:

$$D_{\text{proxy}} = \mathbb{I}(\text{non-centrosymmetric}) \sum_i x_i \left(\frac{Z_i}{100}\right)^4 \cdot \mathbb{I}(Z_i \geq 40) \quad (4)$$

where $\mathbb{I}(\cdot)$ is an indicator function (equal to 1 if the condition holds, 0 otherwise), x_i is the atomic fraction of element i , and Z_i is its atomic number. The fourth-power dependence reflects the relativistic scaling of spin-orbit coupling ($\lambda_{\text{SOC}} \propto Z^4$).

A.3 Crystal Field and Stevens Operator Hamiltonian

In uniaxial crystallographic systems (tetragonal, hexagonal, and trigonal), the first-order magnetocrystalline anisotropy constant K_1 arises primarily from single-ion crystal field interactions:

$$\mathcal{H}_{\text{CF}} = B_2^0 O_2^0 + B_4^0 O_4^0 \quad (5)$$

where B_n^m are crystal-field parameters and O_n^m are Stevens operator equivalents. The second-order operator is defined by:

$$O_2^0 = 3J_z^2 - J(J+1) \quad (6)$$

where J is the total angular momentum quantum number. The sign of the second-order Stevens factor α_J dictates whether the single-ion interaction favors an easy axis ($\alpha_J < 0$, uniaxial anisotropy $K_1 > 0$) or an easy plane ($\alpha_J > 0$).

B MagFormer Architecture Details

This appendix provides the network equations for the MagFormer composition graph transformer described in Section 2.2.

B.1 Node Initialization and Projection

Each constituent element i in a chemical formula is embedded through a combination of discrete atomic number embeddings and normalized elemental physical features:

$$\mathbf{h}_i^{(0)} = \mathbf{W}_{\text{node}} [\mathbf{e}_{Z_i} \parallel x_i \parallel \mathbf{p}_i] \quad (7)$$

where $\mathbf{e}_{Z_i} \in \mathbb{R}^{64}$ is a learned element embedding vector, $x_i \in (0, 1]$ is the atomic fraction, and $\mathbf{p}_i$ is a normalized vector of elemental physical properties (electronegativity, covalent radius, valence, and Stoner parameter).

B.2 Pair-Bias Self-Attention

Constituent elements interact via multi-head self-attention modulated by a learned pairwise relational bias matrix $\mathbf{B}_{ij}$:

$$\mathbf{A}_{ij} = \text{softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_j^T}{\sqrt{d_k}} + \mathbf{B}_{ij} \right) \mathbf{V}_j \quad (8)$$

where query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) projections are computed from $\mathbf{h}_i^{(l)}$. The pairwise bias $\mathbf{B}_{ij}$ is generated by passing elemental feature differences $|\mathbf{p}_i - \mathbf{p}_j|$ through a two-layer multi-layer perceptron.

B.3 Leakage-Free Composition Pooling

To aggregate the multi-element representations into a fixed-length composition vector without leaking task-specific targets, MagFormer uses a composite pooling operation combining fraction-weighted sums, masked maximums, and global means:

$$\mathbf{z}_{\text{comp}} = \left[\sum_{i=1}^N x_i \mathbf{h}_i^{(L)} \parallel \max_{i=1 \dots N} \mathbf{h}_i^{(L)} \parallel \frac{1}{N} \sum_{i=1}^N \mathbf{h}_i^{(L)} \right] \in \mathbb{R}^{384} \quad (9)$$

where L is the final transformer layer and N is the number of elements in the formula. The resulting 384-dimensional representation encodes complex non-linear chemical interactions and is supplied as an input to the tabular ensemble models.